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Fast Initial Rotor Position Estimation Based on Six-Step Voltage Injection and Cost Minimization

Jongwon Choi, Sungjun Oh and Taeyeon Lee

***Abstract*—This paper proposes a fast initial rotor position estimation algorithm for permanent magnet synchronous machines based on six-step operation. Six-step modulation maximizes the inverter voltage utilization and produces one electrical cycle with a single switching action. It generates a large current response within a short duration, enabling rapid rotor position estimation. A cost function based on estimated currents is formulated and analytically investigated using a linear motor model. It exhibits two global minima due to rotor saliency. Interestingly, when magnetic saturation is considered, the cost function structure changes to exhibit a unique global minimum along with a local minimum. It enables rotor position estimation without polarity detection. However, the difference between the global and local minima may become small depending on the rotor position, which may lead to a 180° estimation error. To enhance robustness, two six-step voltage injection method are applied, where the first is applied with no phase shift and the second is phase-shifted by 120°. To identify the global minimum, two methods are employed. First, an approximate initial position is obtained from the current response of the first six-step switching state and used as the initial value for the gradient-based optimization. Second, a two-start gradient-based method is adopted by initializing two points separated by 180° and performing optimization independently from each. The final rotor position is determined by comparing the resulting cost values and selecting the global minimum. The proposed method is experimentally validated to estimate initial rotor position within 4° error in 800–950 μs.**

***Index Terms*—Multi-start methods, gradient descent, optimization method, six-step voltage, pulse injection**

I. Introduction

DRIVEN by the transition toward a low-carbon society ,the demand for electric actuators has increased [1]. High efficiency and power density are required, making interior permanent magnet synchronous machine (IPMSM) promising candidates. The stator current should be oriented and regulated in the rotor flux and accurate rotor angle is required and obtained by position sensors or sensorless algorithms. Rotor position sensors such as encoders, resolvers, and magnetic sensors are widely used, and a calibration process is essential to align the sensor angle with the rotor angle [2]–[4]. The position sensor offset angle is typically calibrated once at the end of line during manufacturing [5]. However, in high-reliability applications such as electric vehicle traction systems and safety-critical industrial servo systems, continuous monitoring of sensor conditions and online calibration methods have become increasingly important to meet functional safety requirements [6]–[8]. Since the position sensor offset angle directly affects the direction and magnitude of the generated torque, real-time monitoring of this offset is essential for reliable motor drive operation. In sensorless operation, accurate initial rotor position information is critical to guarantee maximum torque at startup and to prevent undesired reverse rotation and vibration. Therefore, initial rotor position estimation is critical for both sensorless and sensor-based motor drive systems, where fast and accurate estimation is a key requirement.

A conventional approach is to inject a dc current into the stator before starting to align the rotor with the stator flux [9], [10]. However, the rotor is mechanically forced to move, and the convergence time varies depending on the initial position. Depending on mechanical parameters and loads, the method may take several seconds to converge and induce additional vibrations.

Another approach utilizes rotor magnetic saliency to estimate the initial rotor position using additional voltage injection. Since it does not require mechanical motion, it enables fast estimation times. These methods are classified into continuous high-frequency injection methods [11]–[18] and discontinuous pulse injection methods [19]–[34]. Continuous high-frequency signal injection methods can be classified according to the injection reference frame: the stationary reference frame [11]–[14] and the estimated synchronous reference frame [15]–[18]. To extract the rotor position, digital filtering and demodulation are used. These methods can integrate initial rotor position estimation with sensorless control. However, they require complex signal processing and may have limited convergence rates. Furthermore, continuous signal injection causes additional vibration and acoustic noise, which limits its applicability in noise-sensitive applications.

To reduce complexity and estimation time, pulse voltage injection–based initial rotor position estimation methods have been proposed [19]–[30]. Schmidt et al. [19] first proposed a triple-pulse voltage(TPV) injection method in which positive and negative voltage pulses are sequentially applied to each stator phase. The TPV method applies three pairs of opposite active voltage vectors. For example, in Fig. 1, the positive voltage vectors can be represented by V_1, V_3, and V_5, whereas the corresponding negative voltage vectors are V_4, V_6, and V_2, respectively. These three voltage-vector pairs are sequentially applied, and the incremental current is calculated using the maximum and average currents of each phase. The sector angle and offset angle are then determined for initial rotor position estimation [19], [20]. It does not require a digital filter and

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uses only predefined voltage pulses. The initial rotor position can be estimated using simple arithmetic operations without complex iterative procedures, thereby achieving a consistent and predictable initial position estimation time. However, a pulse voltage injection has a limitation in obtaining an accurate rotor position [21], [23].

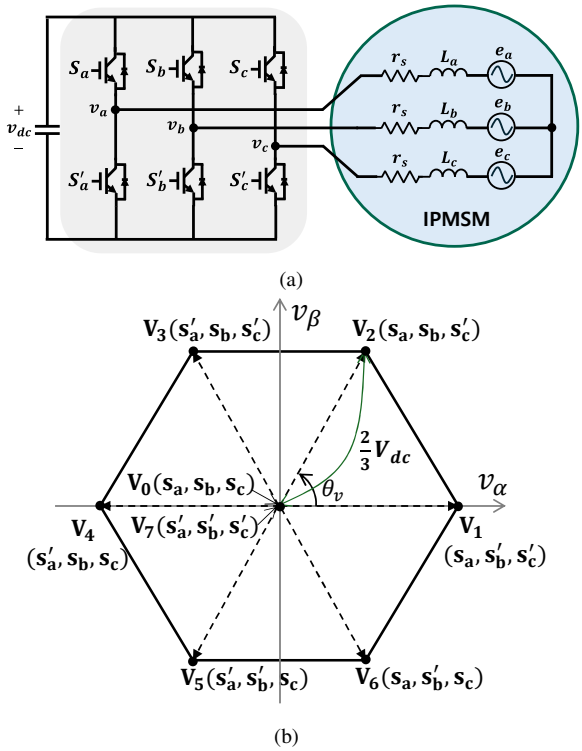


Fig. 1: Inverter motor drive system and voltage vectors: (a) configuration and (b) hexagon of a voltage inverter.

To improve the estimation accuracy, several iteration-based approaches have been proposed [22]–[26]. A three-step initial position estimation method was proposed in [22], where an approximate position is obtained via TPV injection, followed by polarity identification and iterative symmetric voltage injection until convergence. In symmetric voltage injection, the additional voltages are injected to be symmetric to the estimated rotor position. Thus, the injected voltage vectors should be adjusted according to the estimated angle. TPV injection-based estimation was also used for synchronous reluctance motor (SRM) [23]. An approximate rotor position is obtained from current peaks, and then refined by minimizing the error between estimated and measured currents using a PI controller, enabling iterative estimation without additional voltage injection. Although the method achieves high estimation accuracy, its direct application to IPMSMs is limited due to the requirement of the initial flux angle. To reduce the estimation time, a single-pulse voltage injection method in α-axis was proposed in [25], [26]. Unlike TPV injection, which uses three pairs of opposite voltage vectors, single-pulse voltage injection uses only one pair, thereby reducing the number of injected voltage pulses.

Furthermore, unipolar pulse injection methods, in which only positive or negative unipolar voltage is applied, have been proposed [27]–[30]. To improve estimation accuracy, four-step search method was proposed [27] and phase voltage measurement was used [28], [29]. Using two positive voltage injections and one negative voltage injection for polarity identification have also been proposed [30].

Another injection method is line-to-line voltage injection, in which only two phases are excited while the remaining phase is disconnected. Therefore, a series connection is formed between the two phases. Using this configuration, the line-to-line inductances are estimated and used to determine the rotor position [31]. Furthermore, methods based on dc-link voltage dips [32], [33] and terminal voltage measurements [34] have been proposed for rotor position estimation. Although fast estimation within a few milliseconds was achieved, the estimation resolution is relatively low (about 30°) and additional voltage sensing is required.

Previous researches have not achieved fast initial rotor position estimation within $1ms$ while maintaining an error below 5°. One of the main reasons is the voltage injection method, which varies depending on the estimated position. While it could improve estimation accuracy, the sequential execution of voltage injection and estimation causes a bottleneck. Furthermore, these methods do not minimize the signal injection time. Since rotor position is estimated from the current response, maximizing the current response within a short duration is critical for fast estimation.

In this paper, a fast initial rotor position estimation method based on six-step voltage injection is proposed. The robustness and estimation speed are improved by injecting two predefined six-step voltage. This enables parallel execution of the injection and estimation processes. To extract the rotor position, a cost function based on the estimated current is proposed and analytically investigated using linear and nonlinear models. When considering nonlinear motor models, the cost function has a unique global minimum and a local minimum. To search for the global minimum, a gradient descent method is used. Depending on the initial value, this may lead to convergence to a local minimum or slow convergence of the gradient-based optimization. To address this issue, an approximate initial position is obtained from the current response of the first six-step switching state and used as the initial value for the gradient method. Furthermore, to use the inherent 180° shifted between minima, a two-start strategy is employed. Two initial points separated by 180° are used, and the optimization is performed independently from each. The minimizer corresponding to the lower cost is selected as the final estimate. The combination of these two approaches reduces the dependence on the initial value and enables stable and fast convergence even for non-convex cost functions. The effectiveness of the proposed method is validated by experimental results. The main contributions are summarized as follows.

- Two predefined six-step voltages are injected for fast and robust initial rotor position estimation, which allows the injection and estimation to be executed in parallel.
- A cost function based on the estimated current response is formulated and analyzed using both linear and nonlinear motor models.
- A robust gradient-based optimization strategy is developed by combining an approximate initial position

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method and a two-start strategy, improving the convergence speed and reducing the dependence on the initial value.

II. The Motor Current Response

A. Continuous-Time Representation

The IPMSM Model is given as

$$\begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} = r_s \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + \begin{bmatrix} L_{av} - L_{df}\cos 2\theta & -L_{df}\sin 2\theta \\ -L_{df}\sin 2\theta & L_{av} + L_{df}\cos 2\theta \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} - 2\omega L_{df} \begin{bmatrix} -\sin 2\theta & \cos 2\theta \\ \cos 2\theta & \sin 2\theta \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + \omega\psi_m \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix} \tag{1}$$

where i_α, i_β are stator currents in the stationary reference frame [22], v_α, v_β are stator voltages, r_s is stator resistance, ψ_m is permanent magnet (PM) flux linkage coefficient, θ is electric rotor position, and ω is electric rotor speed. L_{av} and L_{df} are defined as the average and differential dq-axis inductances, given by $L_{av} = \frac{L_d+L_q}{2}$, $L_{df} = \frac{L_q-L_d}{2}$ where L_d and L_q are d and q-axis stator inductance. Assuming $\omega = 0$ and high-frequency voltage injection, the inductive impedance is much larger than the resistive impedance of the stator. Therefore, the voltage drop of the stator resistor is negligible, (1) is approximated as,

$$\begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} \approx \begin{bmatrix} L_{av} - L_{df}\cos 2\theta & -L_{df}\sin 2\theta \\ -L_{df}\sin 2\theta & L_{av} + L_{df}\cos 2\theta \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix}. \tag{2}$$

Multiplying both sides of (2) by the inverse matrix, $\begin{bmatrix} L_{av} - L_{df}\cos 2\theta & -L_{df}\sin 2\theta \\ -L_{df}\sin 2\theta & L_{av} + L_{df}\cos 2\theta \end{bmatrix}^{-1}$, it can be obtained as

$$\frac{d}{dt} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \left[\frac{L_{av}}{L_d L_q}\mathbf{I} + \frac{L_{df}}{L_d L_q} \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix} \right] \begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix}, \tag{3}$$

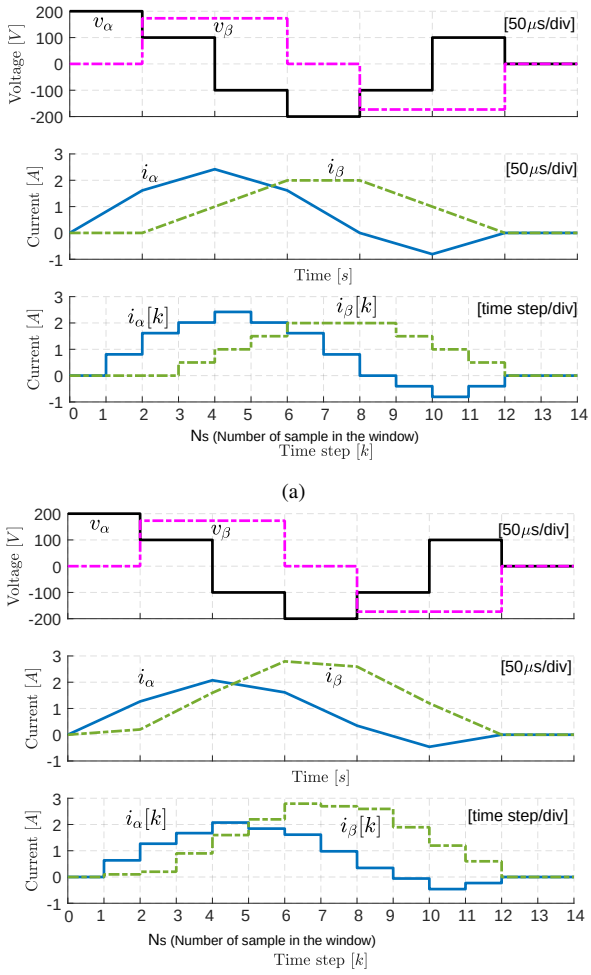
where $\mathbf{I} \in \mathbb{R}^{2\times 2}$ is the identity matrix.

B. Six-Step Voltage

Fig. 1 (a) shows the electric motor drive system, which consists of the three phase half-bridge inverter and IPMSM. The inverter is composed of six switching devices and is capable of generating eight voltage vectors. Fig. 1 (b) shows the voltage limitation hexagon. The vertices of the hexagon are represented by six active vectors. The six active vectors are defined as,

$$\mathbf{V}_{n_v} = \begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} = \frac{2}{3} V_{dc} \begin{bmatrix} \cos(\frac{\pi}{3}(n_v - 1)) \\ \sin(\frac{\pi}{3}(n_v - 1)) \end{bmatrix}, \; n_v \in 1, \ldots, 6, \tag{4}$$

where n_v is the number of vertices and V_{dc} is the dc-link voltage. The zero vector which corresponds to the origin is defined as $\mathbf{V}_0 = \mathbf{V}_7 = \mathbf{0}$. One of the main advantages of the six-step operation is its high voltage utilization [35]. Compared to linear region of space vector modulation, six-step operation enables the inverter to apply the maximum fundamental voltage up to $2V_{dc}/\pi$.



(a)

(b)

Fig. 2: Current response under six-step voltage injection with varying rotor position: (a) $\theta = 0°$ and (b) $\theta = 60°$.

C. Discrete-Time Representation

As shown in Fig. 2, a six-step voltage is utilized to estimate an initial position. The six-step voltage injection uses full duty instead of pulse width modulation. Consequently, it is more suitable to model the system in the discrete-time domain. By applying the forward Euler method, the current derivative term is approximated as $\frac{d}{dt}i \approx \frac{i[k+1]-i[k]}{T_s}$ where T_s is the sampling time. Substituting the approximation result into (3), it follows that

$$\underbrace{\begin{bmatrix} \Delta i_\alpha[k] \\ \Delta i_\beta[k] \end{bmatrix}}_{\Delta \mathbf{i}_{\alpha\beta}[k]} = \frac{T_s}{L_d L_q} \left[L_{av}\mathbf{I} + L_{df} \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix} \right] \underbrace{\begin{bmatrix} v_\alpha[k] \\ v_\beta[k] \end{bmatrix}}_{\mathbf{v}_{\alpha\beta}[k]}, \tag{5}$$

where $\Delta\mathbf{i}_{\alpha\beta}[k] = \mathbf{i}_{\alpha\beta}[k+1] - \mathbf{i}_{\alpha\beta}[k]$ denotes the current increment at the time step k. The current at time step $k+1$ is given by

$$\mathbf{i}_{\alpha\beta}[k+1] = \Delta\mathbf{i}_{\alpha\beta}[k] + \mathbf{i}_{\alpha\beta}[k]. \tag{6}$$

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From (4), each six-step voltages has a magnitude of $2V_{dc}/3$ and points to a vertex of a regular hexagon. The six-step voltage at time step k can be expressed as follows

$$\mathbf{v}_{\alpha\beta}[k] = \frac{2}{3}V_{dc}\begin{bmatrix}\cos(\theta_v[k])\\ \sin(\theta_v[k])\end{bmatrix}, \tag{7}$$

where $\theta_v[k]$ is the angle of the voltage vector at time step k , $\theta_v[k] = \pi/3(n_v[k]-1) = \text{atan2}(v_\beta[k], v_\alpha[k])$. Substituting (7) into (5), the current response at time step k is obtained as

$$\Delta\mathbf{i}_{\alpha\beta}[k] = K_c V_{dc}\left(L_{av}\begin{bmatrix}\cos\theta_v[k]\\ \sin\theta_v[k]\end{bmatrix} + L_{df}\begin{bmatrix}\cos(2\theta-\theta_v[k])\\ \sin(2\theta-\theta_v[k])\end{bmatrix}\right), \tag{8}$$

where $K_c = \frac{2}{3}\frac{T_s}{L_d L_q}$. Note that the incremental current in (8) is determined by two contributing terms. The first term, $L_{av}\begin{bmatrix}\cos\theta_v[k]\\ \sin\theta_v[k]\end{bmatrix}$, represents that the incremental current is aligned with the injected voltage vector and is proportional to the average dq-axis inductance, L_{av}. The other term, $L_{df}\begin{bmatrix}\cos(2\theta-\theta_v[k])\\ \sin(2\theta-\theta_v[k])\end{bmatrix}$, indicates that the incremental current angle is determined by both the rotor position and the injected voltage angle, and its magnitude is proportional to L_{df}.

Assuming zero initial current, i.e, $\mathbf{i}_{\alpha\beta}[0] = \mathbf{0}$, it follows $\mathbf{i}_{\alpha\beta}[1] = \Delta\mathbf{i}_{\alpha\beta}[0]$ from (6). Therefore, the current at time step k can be calculated as follows

$$\mathbf{i}_{\alpha\beta}[k] = \sum_{n_i=0}^{k-1}\Delta\mathbf{i}_{\alpha\beta}[n_i], \tag{9}$$

where n_i is denoted as the sampling number. Fig. 2 shows the $\alpha\beta$-axis currents $\mathbf{i}_{\alpha\beta}$ and corresponding sampled currents $\mathbf{i}_{\alpha\beta}[k]$ when the six-step voltage is applied. Because six distinct switching transitions occur during one electrical period, the injection voltage angle varies six times. Therefore, the sampling frequency f_s should be selected as an integer multiple of six times the injection frequency f_i, i.e., $f_s = 6 \times N_o \times f_i$ where N_o denotes the integer number of samples per switching state. Let N_s denote the total number of samples per cycle. Then, it can be expressed as

$$N_s = N_o \times 6. \tag{10}$$

The current was sampled twice per switch voltage vector, yielding $N_s = 12$ and $N_o = 2$. Note that Fig. 2 (a) and (b) shows the different current responses depending on the rotor position under identical injection voltage conditions. According to the second term in (8), the incremental current varies with the rotor position due to L_{df}, thereby enabling initial rotor position estimation.

III. Initial Rotor Position Estimator with Six-Step Voltage

A. The Cost Function Under Linear Motor Model

From (8) and (9), assuming the motor parameters L_{df} and L_{av} are known and constant and that V_{dc} is measurable, the estimated current becomes a function of $\hat{\theta}$ and can be calculated as,

$$\Delta\hat{\mathbf{i}}_{\alpha\beta}[k] = K_c V_{dc}\left(L_{av}\begin{bmatrix}\cos\theta_v[k]\\ \sin\theta_v[k]\end{bmatrix} + L_{df}\begin{bmatrix}\cos(2\hat{\theta}-\theta_v[k])\\ \sin(2\hat{\theta}-\theta_v[k])\end{bmatrix}\right), \tag{11}$$

$$\hat{\mathbf{i}}_{\alpha\beta}[k] = \sum_{n_i=0}^{k-1}\Delta\hat{\mathbf{i}}_{\alpha\beta}[n_i], \tag{12}$$

where $\hat{\mathbf{i}}_{\alpha\beta}[k]$ denotes the estimated current at time step k. Using (9) and (12), the difference between estimated and real current can be calculated

$$\hat{\mathbf{i}}_{\alpha\beta}[k] - \mathbf{i}_{\alpha\beta}[k] = \sum_{n_i=0}^{k-1}\left(\Delta\hat{\mathbf{i}}_{\alpha\beta}[n_i] - \Delta\mathbf{i}_{\alpha\beta}[n_i]\right). \tag{13}$$

By subtracting (8) from (11) and applying the trigonometric sum-to-product formulas, the incremental current difference between the real and estimated values can be derived as

$$\begin{aligned}&\Delta\hat{\mathbf{i}}_{\alpha\beta}[k] - \Delta\mathbf{i}_{\alpha\beta}[k]\\ &= 2K_c V_{dc} L_{df}\sin\Delta\theta\begin{bmatrix}-\sin(2\theta+\Delta\theta-\theta_v[k])\\ \cos(2\theta+\Delta\theta-\theta_v[k])\end{bmatrix},\end{aligned} \tag{14}$$

where $\Delta\theta$ is the estimation error, defined as $\Delta\theta = \hat{\theta} - \theta$, $\Delta\theta \in [0, 2\pi)$. Note that the average dq inductance, L_{av}, is completely eliminated in (14) and the current difference is proportional L_{df}. Thus, the robustness of the initial angle estimate depends on L_{df}. By taking the 2-norm on both sides of (13) and substituting (14), it follows that

$$\begin{aligned}\|\hat{\mathbf{i}}_{\alpha\beta}[k] - \mathbf{i}_{\alpha\beta}[k]\|^2 &= 4K_c^2 V_{dc}^2 L_{df}^2\sin^2\Delta\theta\\ &\left\|\sum_{n_i=0}^{k-1}\begin{bmatrix}-\sin(2\theta+\Delta\theta-\theta_v[n_i])\\ \cos(2\theta+\Delta\theta-\theta_v[n_i])\end{bmatrix}\right\|^2,\end{aligned} \tag{15}$$

where $\|\cdot\|$ is 2-norm operator. Intuitively, $\|\hat{\mathbf{i}}_{\alpha\beta}[k] - \mathbf{i}_{\alpha\beta}[k]\|$ can be represented by angle difference which is $\Delta\theta$. Based on the idea, the cost function is constructed as the summation of the differences between the estimated and measured currents over a window size of N_s and is defined as

$$J = \sum_{k=1}^{N_s}\|\hat{\mathbf{i}}_{\alpha\beta}[k] - \mathbf{i}_{\alpha\beta}[k]\|^2. \tag{16}$$

Substituting (15) into (16), the cost function is derived as,

$$\begin{aligned}J &= K_j\sin^2\Delta\theta\underbrace{\sum_{k=1}^{N_s}\left\|\sum_{n_i=0}^{k-1}\begin{bmatrix}-\sin(2\theta+\Delta\theta-\theta_v[n_i])\\ \cos(2\theta+\Delta\theta-\theta_v[n_i])\end{bmatrix}\right\|^2}_{K_t}\\ &= K_j\,K_t\sin^2\Delta\theta,\end{aligned} \tag{17}$$

where $K_j = 4K_c^2V_{dc}^2L_{df}^2$, $K_t = \sum_{k=1}^{N_s}(a_{k_s}^2 + b_{k_s}^2)$,

$$a_{k_s} = \sum_{n_i=0}^{k-1}\cos\left(\left\lfloor\frac{n_i}{N_o}\right\rfloor\frac{\pi}{3}\right), \text{ and}$$

$$b_{k_s} = \sum_{k=0}^{k-1}\sin\left(\left\lfloor\frac{n_i}{N_o}\right\rfloor\frac{\pi}{3}\right).$$

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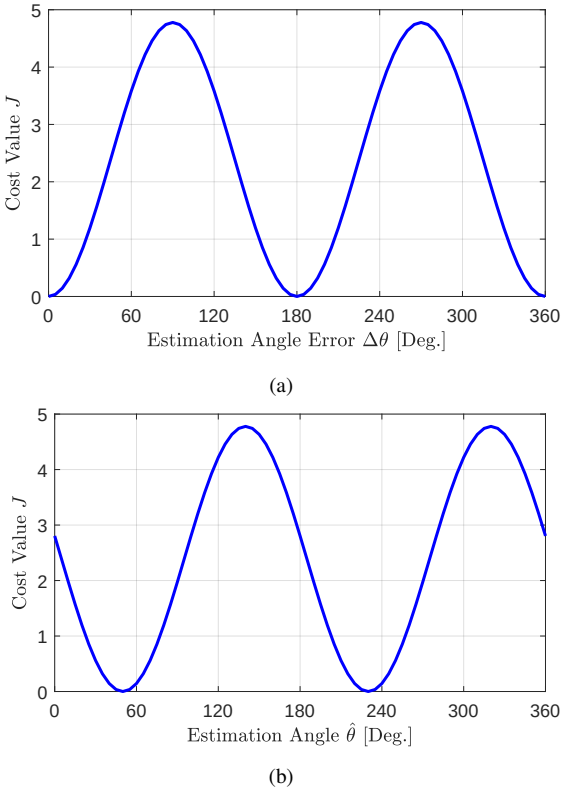


Fig. 3: Cost function under the linear motor model: (a) function of the estimation angle error $\Delta\theta$ and (b) function of the estimation angle $\hat{\theta}$.

$\lfloor\cdot\rfloor$ denotes the floor function, which returns the greatest integer less than or equal to the input variable. K_t is determined by the window size N_s. For example, $K_t = 90$ when $N_s = 12$, and $K_t = 12$ when $N_s = 6$. The detailed derivation of K_t is provided in the Appendix. Note that the cost function in (17) is formed by the terms proportional $\sin^2\Delta\theta$. Fig 3 shows the cost function of $\Delta\theta$ and $\hat{\theta}$. The nominal parameters listed in Table I are used, and the window size is set to $N_s = 12$. In

TABLE I: Nominal Parameters of an IPMSM and inverter used in simulations and experiments.

Stator resistance (R)	0.39 Ω
d axis inductance (L_d)	6.25 mH
q axis inductance (L_q)	8.68 mH
PM flux linkage constant (ψ_m)	0.11 Wb
Numbers of poles (P)	6 poles
DC-link voltage (V_{dc})	300 V
Rated current(I_s)	8.7Arms

Fig 3 (a), the cost function has two minimum points at 0° and 180° due to the rotor saliency. Furthermore, substituting $\Delta\theta = \hat{\theta} - \theta$ into (17), it follows that

$$J = K_j\, K_t \sin^2(\hat{\theta} - \theta). \tag{18}$$

This cost function is obtained by shifting the function in (17) by θ. Fig 3 (b) shows the cost function when θ = 50°. This cost function has two global minima at 50° and 230° due to the inherent rotor symmetry. To distinguish between these two minima, an additional polarity-detection method is

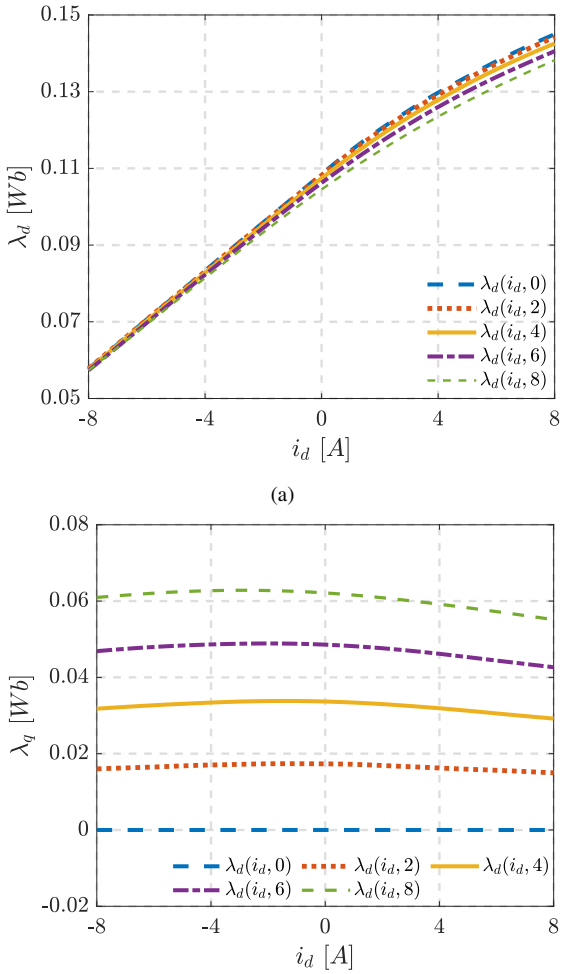


Fig. 4: The effect of dq-axis current variation on dq flux linkages: (a) d-axis flux linkage and (b) q-axis flux linkage.

theoretically required. However, by incorporating the nonlinear motor model into the proposed algorithm, the initial rotor angle can be estimated without an additional polarity detection algorithm.

B. Nonlinear Motor Model

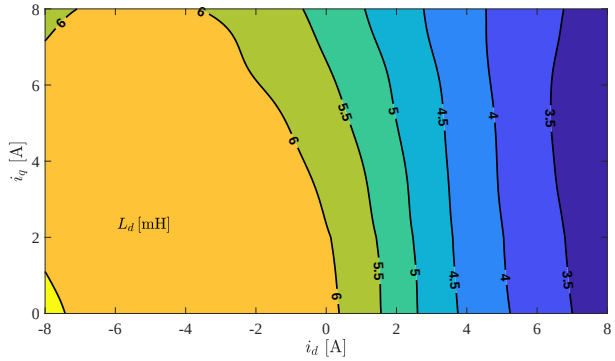
Fig. 4 shows dq-axis flux linkage $\lambda_d(i_d, i_q)$ and $\lambda_q(i_d, i_q)$ as functions of dq-axis current. The flux data were obtained experimentally [36]. The d-axis flux linkage exhibits linear behavior for negative d-axis current, but becomes nonlinear for positive d-axis current in Fig. 4 (a), due to magnetic saturation. Fig. 4 (b) shows that the q-axis flux linkage decreases more for $i_d > 0$, while showing weak sensitivity in the negative d-axis current region. Fig. 5 shows the contour plots of the inductances in the dq-axis current plane. The inductances were calculated from the flux map data shown in Fig. 4 as follows:

$$L_d(i_d, i_q) = \frac{\partial\lambda_d}{\partial i_d} \approx \frac{\lambda_d(i_d + \Delta i_d, i_q) - \lambda_d(i_d, i_q)}{\Delta i_d}, \tag{19}$$

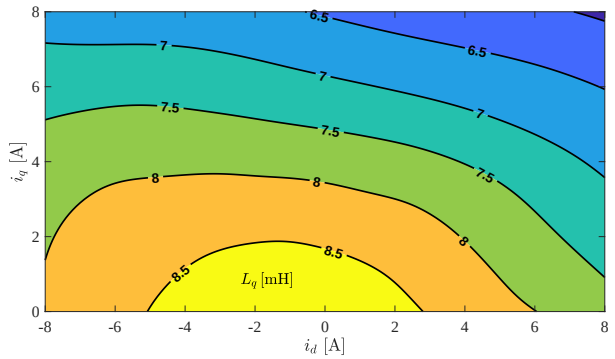
$$L_q(i_d, i_q) = \frac{\partial\lambda_q}{\partial i_q} \approx \frac{\lambda_q(i_d, i_q + \Delta i_q) - \lambda_q(i_d, i_q)}{\Delta i_q}, \tag{20}$$

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(a)



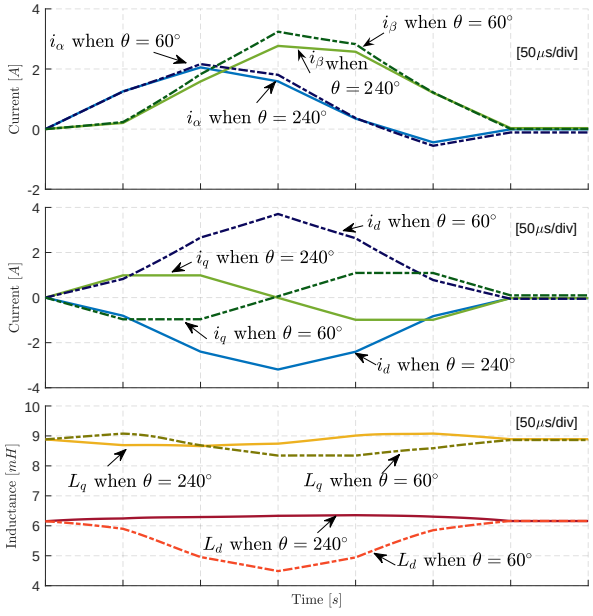
(b)

Fig. 5: Inductance contours for i_d and i_q: (a) L_d and (b) L_q.

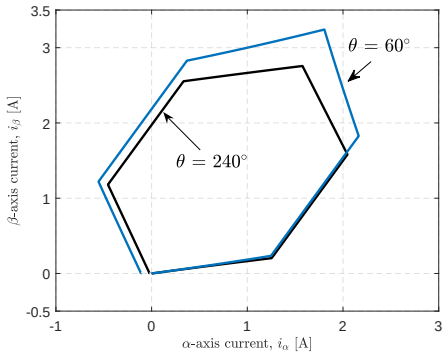
where L_d and L_q is the incremental inductance [37]. In Fig. 5, the dq-axis inductances are not constant but vary with the operating point in the dq-axis current plane. The current-dependent variation of the dq-axis inductances can affect the parameter sensitivity of PMSM motor control and sensorless estimation algorithms [39], [40]. Fig. 5 (a) shows that L_d is almost constant about 6 mH for negative d-axis currents. However, when $i_d > 0$, L_d decreases monotonically as the d-axis current increases. Fig. 5 (b) shows L_q decreases steeply with increasing i_q. Moreover, as the magnitude of i_d increases, L_d also decreases slightly. L_d varies significantly depending on the direction of the d-axis current. In contrast, L_q significantly depends on the direction of the d-axis current. Consequently, the magnitude and direction of d-axis current strongly affect L_d, which enables the proposed method to resolve the absolute polarity determination problem.

C. The Cost Function Under the Nonlinear Motor Model

Considering the inductance variation caused by magnetic saturation, $L_{av}(i_d, i_q)$ and $L_{df}(i_d, i_q)$ depend on the dq-axis currents. Since i_d and i_q are functions of the stator currents and the rotor position, they can be equivalently represented as $L_{av}(i_\alpha, i_\beta, \theta)$ and $L_{df}(i_\alpha, i_\beta, \theta)$. For notational simplicity, the nonlinear inductances are defined as $L_{av}[k] = L_{av}(i_\alpha[k], i_\beta[k], \theta)$ and $L_{df}[k] = L_{df}(i_\alpha[k], i_\beta[k], \theta)$. Similarly, the inductances evaluated at the estimated rotor position are defined as $\hat{L}_{av}[k] = L_{av}(i_\alpha[k], i_\beta[k], \hat{\theta})$ and $\hat{L}_{df}[k] = L_{df}(i_\alpha[k], i_\beta[k], \hat{\theta})$. Then, the variation of the nonlinear inductances can be expressed as



(a)



(b)

Fig. 6: Current response under nonlinear motor model with six-step voltage injection for $\theta = 60°$ and $\theta = 240°$: (a) $\alpha\beta$ and dq-axis currents and variation of dq-axis inductance and (b) Lissajous curve of i_α- i_β.

$$\hat{L}_{df}[k] = L_{df}[k] + \kappa_{df}[k]\Delta\theta, \tag{21}$$

$$\hat{L}_{av}[k] = L_{av}[k] + \kappa_{av}[k]\Delta\theta, \tag{22}$$

where $\kappa_{df}[k] = (\hat{L}_{df}[k] - L_{df}[k])/\Delta\theta$ and $\kappa_{av}[k] = (\hat{L}_{av}[k] - L_{av}[k])/\Delta\theta$. To account for the nonlinear inductance variation, substituting (21) and (22) into (11), the incremental current difference can be expressed as

$$\Delta\hat{\mathbf{i}}_{\alpha\beta}[k] - \Delta\mathbf{i}_{\alpha\beta}[k] = 2K_c V_{dc}\left(\Delta\theta\,\mathbf{e}_{st}[k] + \sin\Delta\theta\,\mathbf{e}_{li}[k]\right), \tag{23}$$

where $\mathbf{e}_{st}[k]$ denotes the saturation-induced component resulting from the nonlinear inductance variation, whereas $\mathbf{e}_{li}[k]$

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denotes the saliency-related component. They are defined as

$$\mathbf{e}_{st}[k] = \frac{\kappa_{av}[k]}{2}\begin{bmatrix}\cos\theta_v[k]\\ \sin\theta_v[k]\end{bmatrix} + \frac{\kappa_{df}[k]}{2}\begin{bmatrix}\cos(2\theta+\Delta\theta-\theta_v[k])\\ \sin(2\theta+\Delta\theta-\theta_v[k])\end{bmatrix},$$
$$\mathbf{e}_{li}[k] = L_{df}[k]\begin{bmatrix}-\sin(2\theta+\Delta\theta-\theta_v[k])\\ \cos(2\theta+\Delta\theta-\theta_v[k])\end{bmatrix}.$$

Using (23), (13), and (16), the cost function considering the motor nonlinearity is obtained as

$$\begin{aligned} J &= K_j\sum_{k=1}^{N_s}\left\|\sum_{n_i=0}^{k-1}\Delta\theta\mathbf{e}_{st}[n_i] + \sum_{n_i=0}^{k-1}\sin\Delta\theta\mathbf{e}_{li}[n_i]\right\|^2 \\ &= K_j\sum_{k=1}^{N_s}\left\{\Delta\theta^2\left\|\sum_{n_i=0}^{k-1}\mathbf{e}_{st}[n_i]\right\|^2 + \sin^2\Delta\theta\left\|\sum_{n_i=0}^{k-1}\mathbf{e}_{li}[n_i]\right\|^2\right. \\ &\quad \left. +2\Delta\theta\sin\Delta\theta\left(\sum_{n_i=0}^{k-1}\mathbf{e}_{st}[n_i]\right)^T\left(\sum_{n_i=0}^{k-1}\mathbf{e}_{li}[n_i]\right)\right\}. \end{aligned} \tag{24}$$

The key difference between (17) and (24) is the presence of the saturation-induced term $\Delta\theta^2\left\|\sum_{n_i=0}^{k-1}\mathbf{e}_{st}[n_i]\right\|^2$. In the linear model of (17), the cost function becomes zero at $\Delta\theta = \pi$ because the saliency-related term is proportional to $\sin^2\Delta\theta$. However, in (24), when $\Delta\theta = \pi$, the saliency-related term and the cross term vanish because $\sin\pi = 0$, and the cost function does not become zero and is given by $J|_{\Delta\theta=\pi} = K_j\pi^2\sum_{k=1}^{N_s}\left\|\sum_{n_i=0}^{k-1}\mathbf{e}_{st}[n_i]\right\|^2$. This term breaks the 180° symmetry of the cost function, ensuring that the cost function has a unique global minimum. The magnitude of this term is determined by the nonlinear inductance sensitivity coefficients κ_{av} and κ_{df}, which become pronounced when a sufficiently large positive d-axis current response is generated.

The saliency-related component $\mathbf{e}_{li}[k]$ is proportional to $L_{df}[k]$, which represents the saliency level of the motor. A larger L_{df} increases the slope and contrast of the cost function, thereby improving the robustness against measurement noise. On the other hand, the saturation-induced component $\mathbf{e}_{st}[k]$ is determined by the nonlinear inductance sensitivity coefficients $\kappa_{av}[k]$ and $\kappa_{df}[k]$. The larger the value of this coefficient, the stronger the saturation-induced term becomes, and the larger the cost function difference between $J(\theta)$ and $J(\theta+\pi)$. Therefore, sufficient saliency and magnetic saturation are important for achieving robust initial rotor position estimation. The proposed method can be applied to surface-mounted PMSM (SPMSM). However, due to the weak saliency of SPMSM, the pulse duration may need to be increased. This may increase the estimation time and torque oscillation.

Fig. 6 shows the current responses corresponding to different rotor angle under nonlinear flux model. Since 60° and 240° are separated by 180°, the same current responses are expected from (3) under the linear motor model. Fig. 6 (b) shows the Lissajous curves of $\alpha\beta$-axis currents, which exhibit distinct current responses. When the rotor angle is 60°, d-axis current increases in the positive direction, whereas at 240° d-axis current increases in the negative direction. As a result, the opposite directions of d-axis current cause different d-axis inductance variations. Therefore, magnetic saturation causes a difference in the current magnitude on the Lissajous curve. At 60°, magnetic saturation reduces the effective d-axis inductance, resulting in a larger current response, whereas at 240°, the relatively larger d-axis inductance produces a smaller current response. Consequently, the polarity difference between the two rotor positions separated by 180° can be identified from the magnitude of the positive d-axis current response.

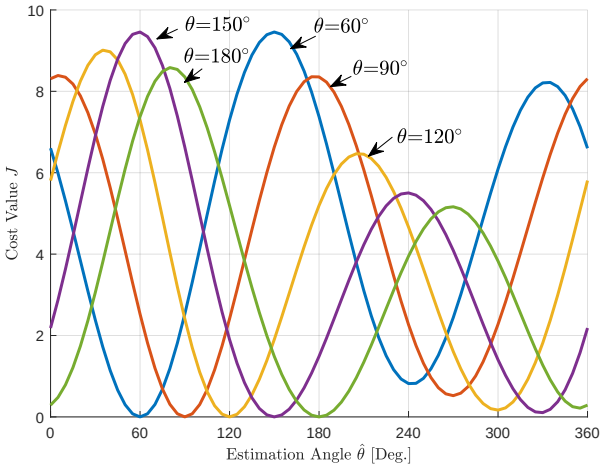


Fig. 7: Cost functions under a nonlinear model with six-step voltage injection without phase shift for rotor angles of 60°, 90°, 120°, 150°, and 180°.

D. Cost Function Variation with Six-Step Voltage Injection Phase and Rotor Position

A single six-step voltage injection can be used to identify the absolute rotor position; however, the robustness of the estimation may depend on the actual rotor position. Fig. 7 shows the cost functions obtained from the nonlinear motor model for rotor angles ranging from 60° to 180°, with an interval of 30°. The minimum value of cost function is achieved when the estimated angle coincides with the rotor angle. The cost function has a local minimum at the point where the estimated angle differs from the rotor angle by 180°. For example, when θ = 60°, the cost function has a global minimum at $\hat{\theta} = 60°$ and a local minimum at $\hat{\theta} = 240°$. Therefore, by considering the difference between the local minimum and the global minimum, the rotor angle can be accurately estimated. However, as the rotor angle varies, the difference between the global and local minima can decrease. In particular, when $\theta = 150°$, this difference becomes the smallest, as shown in Fig. 7, which may degrade the robustness of the estimation. Based on (4), the six-step voltage is transformed into the rotating dq reference frame. Then, the corresponding d-axis current response is obtained as

$$i_d[k] = \frac{2}{3}V_{dc}T_s\frac{1}{L_d}\sum_{n_i=0}^{k-1}\cos(\theta_v[n_i]-\theta) \tag{25}$$

where $i_d[k]$ is d-axis current at k step. When the rotor position is shifted by 180°, the corresponding d-axis current response

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is reversed. Therefore, to distinguish between the two rotor positions separated by 180°, the maximum absolute value of the d-axis current during one six-step voltage injection period can be obtained as

$$|i_d|_{max} = \frac{2}{3} V_{dc} T_s \frac{N_o}{L_d} \max_{k=0,\cdots,5} |\sum_{n_i=0}^{k-1} \cos(\frac{\pi}{6} n_i - \theta)| \quad (26)$$

For normalized representation, the maximum d-axis current response is expressed as

$$|i'_d|_{\max} = \frac{|i_d|_{\max}}{\frac{2}{3} V_{dc} T_s \frac{N_o}{L_d}}.$$

Fig. 8 shows the normalized maximum d-axis current response, $|i'_d|_{\max}$, for the six-step voltage injections with 0° and 120° phase shifts. For the 0° phase-shifted injection, $|i'_d|_{\max}$ reaches its maximum value at a rotor position of 60°, whereas it reaches its minimum value at 150°. For the 120° phase-shifted injection, $|i'_d|_{\max}$ has its maximum values at rotor positions of 0° and 180°, and its minimum value at 90°. Therefore, the combined use of the 0° and 120° six-step voltage injections ensures a sufficiently large d-axis current response over the entire rotor position range.

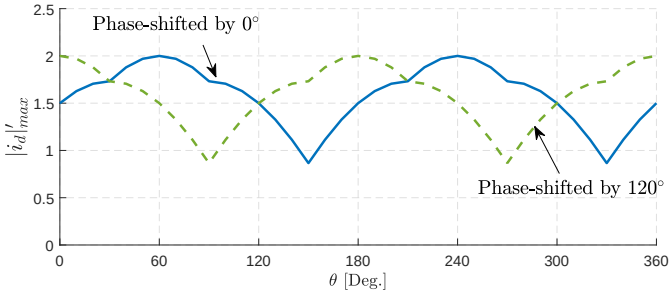


Fig. 8: Effect of rotor position on the normalized maximum d-axis current response, $|i'_d|_{\max}$, under the 0° and 120° six-step voltage injections.

Fig. 9 shows the current responses for two six-step voltage injection cases: without phase shift and with a 120° phase shift, when $\theta = 150°$ and $\theta = 330°$. Fig 9 (a) shows the six-step voltage is applied sequentially from n_v from 1 to 6, which is defined as the case without phase shift. In this case, the variation of the d-axis current is smaller than the variation shown in Fig. 6 (a). This results in a reduced variation of L_d, thereby decreasing the difference between $\alpha\beta$-axis current responses at $\theta = 150°$ and $\theta = 330°$. Consequently, the difference between the global and local minimum of the cost function is reduced. Fig. 9 (b) shows the six-step voltage applied following the order $n_v = 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 1 \rightarrow 2$, which is defined as a six-step voltage with a 120° phase shift. Under the 120° phase shift six-step, i_d reaches 3 A at $\theta = 150°$, while it reaches -3 A at $\theta = 330°$. The difference leads to distinct $\alpha\beta$-axis current responses in Fig. 9 (b). This result is consistent with the analytical result shown in Fig. 8. At $\theta = 150°$ or 330°, the 120° phase shift gives twice the $|i'_d|_{\max}$ of the 0° phase shift. Therefore, at $\theta = 150°$ or 330°,

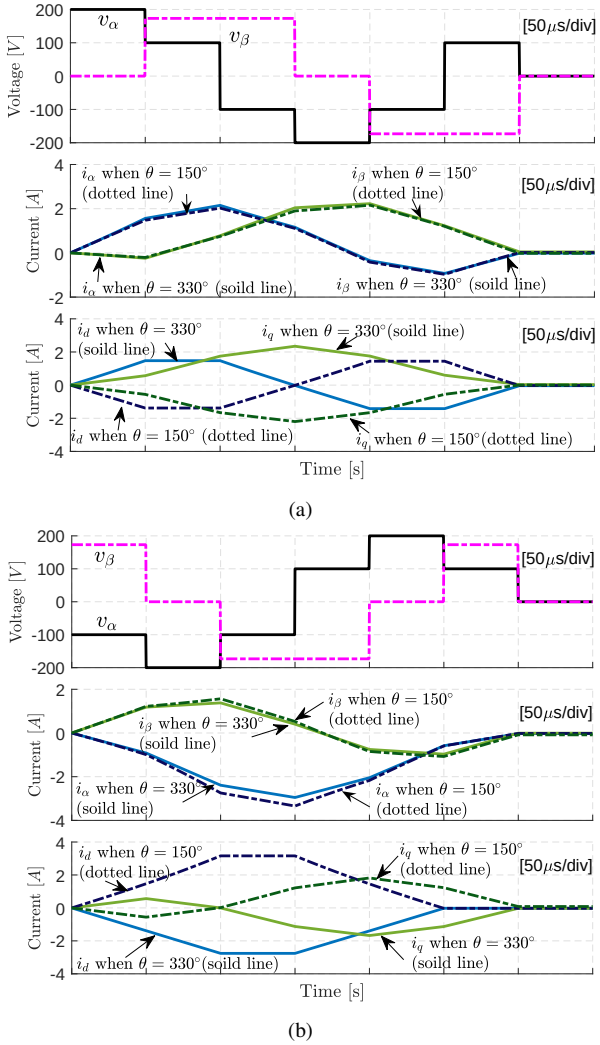


Fig. 9: Current response by six-step voltage when θ = 150° and θ = 330°: (a) without phase shift and (b) with a 120° phase shift.

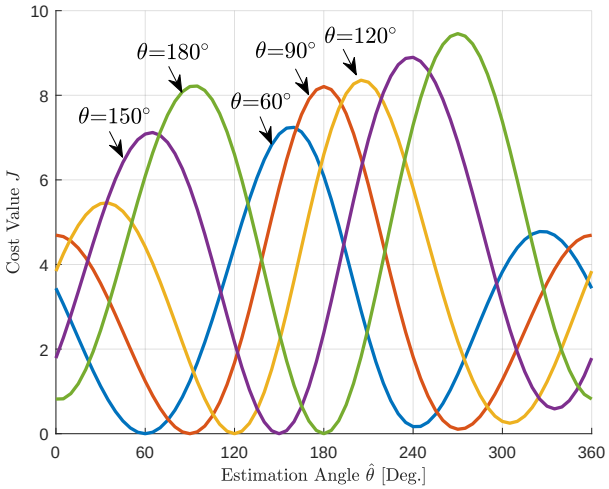


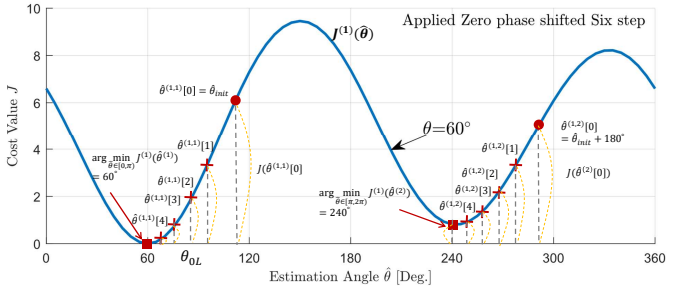
Fig. 10: Cost functions under a nonlinear model with six-step voltage injection phase-shifted by 120° for rotor angles of 60°, 90°, 120°, 150°, and 180°.

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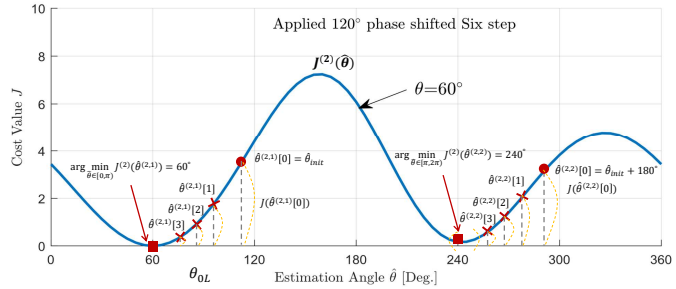
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the 120° phase-shifted injection provides clearer information for resolving the 180° polarity ambiguity.

Fig. 10 shows the cost function obtained with six-step voltage injection with a 120° phase shift. Compared with Fig. 7, a larger difference between the global and the local minima is observed when $\theta = 150°$, indicating improved robustness in initial angle estimation near 150° and 330°. In contrast, at a rotor angle of 60°, six-step voltage injection with a 120° phase shift reduces the difference between the global and local minima. The effectiveness of the applied six-step voltage phase depends on the initial rotor angle. Therefore, two six-step voltages with different phase shifts are sequentially applied, and their corresponding results are analyzed to enhance the initial angle estimation performance..



(a)



(b)

Fig. 11: Gradient method depending on six-step voltage injection: (a) without phase shift and (b) with a 120° phase shift.

IV. The Estimation Method Based on the Cost Function

To clearly distinguish between the global and local minima, the six-step voltage injection is applied twice, resulting in two independent cost functions, which are defined by $J^{(1)}$ and $J^{(2)}$. The cost function $J^{(1)}$ and $J^{(2)}$ are obtained using six-step voltage injection without phase shift and with a 120° phase shift, respectively. The convergence analysis assumes a smooth non-convex cost function with two 180° separated local minimizers and no other local minimizers in the considered electrical angle range. Fig. 11 shows two cost functions corresponding to different six-step voltage phases when $\theta = 60°$. In both cases, the global minimizer occurs at 60° and the local minimizer occures at 240°. But the gap between the global and local minimizers varies between the two cost functions. To ensure robustness of angle estimation, a gap between the global and local minimizers should be sufficiently large. Consequently, the estimated rotor angle is determined using the cost function that provides the largest gap between the global and local minimizers. The cost function index $k_j^\star$ is selected as,

$$k_j^* = \arg\max_{k_j \in \{1,2\}} \left| \min_{\hat{\theta}^{(1)} \in [0,\pi)} J^{(k_j)}(\hat{\theta}^{(1)}) - \min_{\hat{\theta}^{(2)} \in [\pi,2\pi)} J^{(k_j)}(\hat{\theta}^{(2)}) \right|, \tag{27}$$

where $\arg\min(\cdot)$ denotes the argument that minimizes the input function and $\min(\cdot)$ denotes the minimum value of the input function. Note that the search domain of the cost function is divided into two angular regions to distinguish two minimizers. The term $\min_{\hat{\theta}\in[0,\pi)} J^{(k_j)}(\hat{\theta})$ denotes the minimum cost value obtained over the interval $\hat{\theta} \in [0, \pi)$. Then, the final estimated rotor angle is obtained by

$$\hat{\theta}_{\text{final}} = \arg\min_{\hat{\theta}\in[0,2\pi)} J^{(k_j^*)}(\hat{\theta}). \tag{28}$$

To obtain the solution for (27) and (28), an optimization method is required to find the minimum value of cost function. In this study, a gradient descent algorithm is used because of its computational efficiency. However, it may suffer slow convergence and congverge to an undesired local minimizer depending on the initial point. To overcome these limitation, an approximate inital angle estimation algorithm and two-start gradient-based method are proposed.

Algorithm 1 Approximate Estimation Determination

Require: Measured $\mathbf{i}_{\alpha\beta}[N_o]$ under six-step voltage injection without phase shift

```
Initialize ‖ζ‖_max ← 0 and θ̂_init ← 0
for θ̂_ran = 0° to 180° do
    Compute ‖ζ(θ̂_ran)‖ using (31)
    if ‖ζ(θ̂_ran)‖ > ‖ζ‖_max then
        ‖ζ‖_max ← ‖ζ(θ̂_ran)‖
        Compute θ̂_init using (32)
    end if
end for
return θ̂_init
```

A. An Approximate initial angle estimation algorithm

According to (10), under six-step injection without phase shift, the same voltage is injected for N_o consecutive intervals the voltage vector angles are given by $\theta_v[0], \ldots, \theta_v[N_o] = 0°$. Therefore, from (8) and (9), the current at time step N_o can be expressed as

$$\mathbf{i}_{\alpha\beta}[N_o] = K_c V_{dc} N_o \left(L_{av} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + L_{df} \begin{bmatrix} \cos 2\theta \\ \sin 2\theta \end{bmatrix} \right). \tag{29}$$

Then, the estimated current corresponding to an arbitrary estimated angle $\hat{\theta}_{ran}$ can defined as

$$\hat{\mathbf{i}}_{\alpha\beta}[N_o] = K_c V_{dc} N_o \left(L_{av} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + L_{df} \begin{bmatrix} \cos 2\hat{\theta}_{ran} \\ \sin 2\hat{\theta}_{ran} \end{bmatrix} \right), \tag{30}$$

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where $\hat{\theta}_{ran}$ is arbitrarily chosen within $\hat{\theta}_{\text{ran}} \in [0, \pi)$. The vector $\boldsymbol{\zeta}$ is constructed by subtracting the estimated current from the measured current in the α-axis and adding them in the β-axis at time step N_o. Using (29) and (30), it follows that

$$\boldsymbol{\zeta} = \begin{bmatrix} \zeta_\alpha \\ \zeta_\beta \end{bmatrix} = \begin{bmatrix} \hat{i}_\alpha[N_o] - i_\alpha[N_o] \\ \hat{i}_\beta[N_o] + i_\beta[N_o] \end{bmatrix} = \underbrace{2K_c V_{dc} N_0 L_{df} \sin(\theta + \hat{\theta}_{ran})}_{\|\boldsymbol{\zeta}\|} \begin{bmatrix} -\sin(\hat{\theta}_{ran} - \theta) \\ \cos(\hat{\theta}_{ran} - \theta) \end{bmatrix}. \quad (31)$$

Then, an approximate initial angle estimate can be obtained

$$\hat{\theta}_{init} = \hat{\theta}_{ran} + \text{atan2}(\zeta_\alpha, \zeta_\beta), \quad (32)$$

where $\hat{\theta}_{init}$ is used as the initial starting point for the gradient-based algorithm. A sufficiently large magnitude of $\|\boldsymbol{\zeta}\|$ is required to ensure reliable initial angle estimation. When $\theta = 90°$ and $\hat{\theta}_{ran} = 90°$, $\|\boldsymbol{\zeta}\|$ becomes zero, making it difficult to distinguish the angle of the rotor. Therefore, Algorithm 1 is proposed as follows.

B. Two-Start Gradient-Based Optimization Method for Initial Angle Estimation

A multi-start gradient-based method has been proposed to handle optimization problems with multiple local minima [38]. Since the cost function exhibits two minima, as shown in Figs. 7 and 10, a two-start gradient-based method is employed. In this approach, an approximate initial angle estimate obtained from (32) is required, and the six-step voltage injection is performed in advance. Two initial points are selected by using the inherent 180° phase difference between the global and local minimizers. Then, gradient descent is independently performed from each initial point to minimize the corresponding cost function. By comparing the resulting cost values from the two optimization paths, the difference the global and local minima can be evaluated. The cost function with the maximum difference is selected, and the angle corresponding to its minimum value is determined as the final rotor position estimate. The overall method is detailed in Algorithm 2.

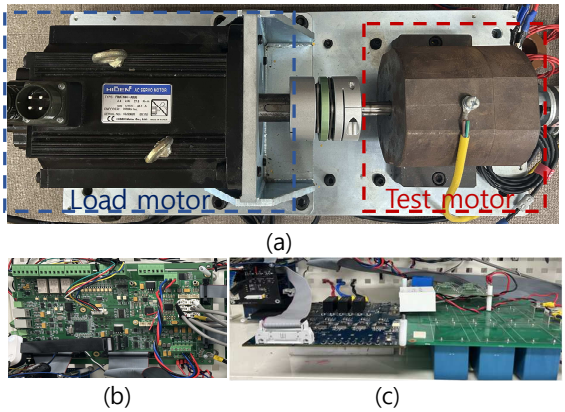


Fig. 12: Experimental environment: (a) Dynamo system, control board, and test inverter.

Algorithm 2 Two-Start Gradient-Based Initial Angle Estimation

Require: Approximate estimate $\hat{\theta}_{\text{init}}$ from Algorithm 1
Require: Cost functions $J^{(k_j)}$ computed using N_s samples where $k_j = 1$ denotes no phase shift and $k_j = 2$ denotes a 120° phase shift
Initialize candidate set:
$\hat{\theta}^{(k_j,1)} \leftarrow \hat{\theta}_{init}$ and $\hat{\theta}^{(k_j,2)} \leftarrow \hat{\theta}_{init} + \pi$
for $i = 1$ to 2 **do**
  while $|\nabla J^{(k_j)}(\hat{\theta}^{(k_j,i)})| > \epsilon$ **do**
    Compute gradient $\nabla J^{(k_j)}(\hat{\theta}^{(k_j,i)})$
    Update $\hat{\theta}^{(k_j,i)} \leftarrow \hat{\theta}^{(k_j,i)} - \gamma \nabla J^{(k_j)}(\hat{\theta}^{(k_j,i)})$
  end while
  $J_{\min}^{(k_j,i)} \leftarrow J^{(k_j)}(\hat{\theta}^{(k_j,i)})$
end for
Selection:
After completing both six-step voltage injections, determine the final rotor angle according to (27) and (28)

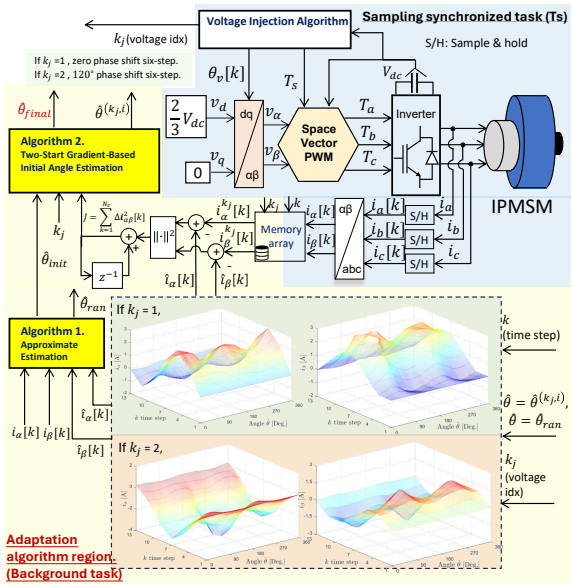


Fig. 13: Block diagram of the proposed method.

V. EXPERIMENTAL RESULTS

Fig. 12 shows the IPMSM, control board, and inverter used in the experiment. The control software was implemented on a TMS28377D digital signal processor, and a dc-link voltage is applied to 300 V. The period of six-step was set to 300μs. The stator currents were sampled every 25 μs, and the number of samples for each switching vector, N_o, was set to 2. A longer six-step voltage injection duration increases the phase current magnitude, which can enhance the robustness of the proposed algorithm. However, an excessively long injection duration may increase the total estimation time and generate additional oscillatory torque.

Fig. 13 shows the overall configuration of the proposed observer. The proposed algorithm consists of two main parts: the voltage injection/current acquisition part and the adaptation algorithm part. In the voltage injection/current acquisition part,

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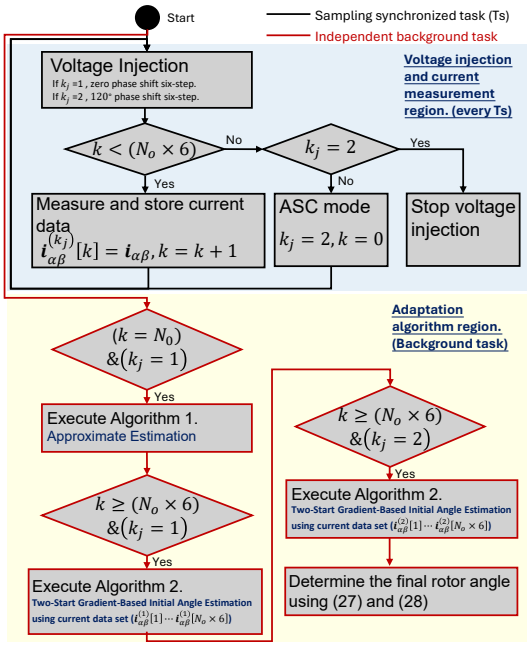


Fig. 14: Flowchart of the proposed algorithm

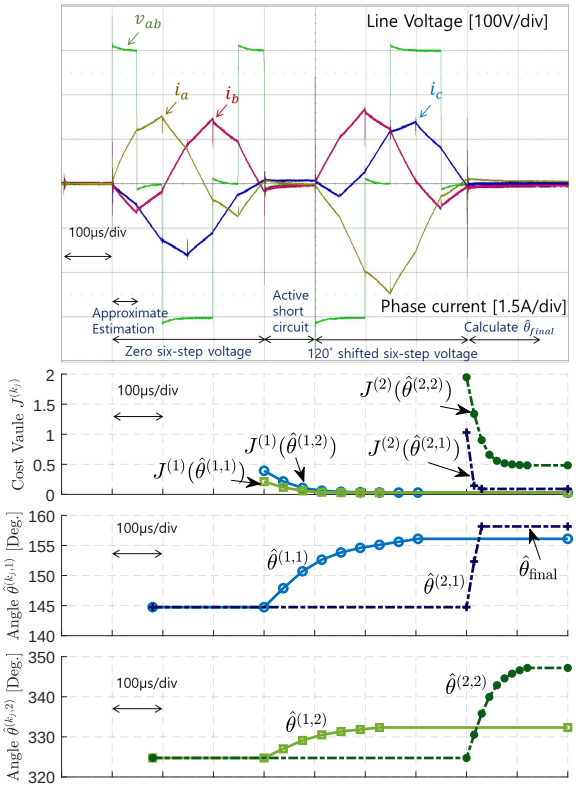


Fig. 15: The current response, six-step voltage injections, and cost value, and estimated angle when $\theta = 155.4°$.

the voltage generation block produces the voltage angle and the phase index, k_j. The stator currents are then measured and stored at each sampling index. The second part is the adaptation algorithm, which determines the final estimated rotor position by minimizing the proposed cost function. In this part, the estimated current response is obtained from the

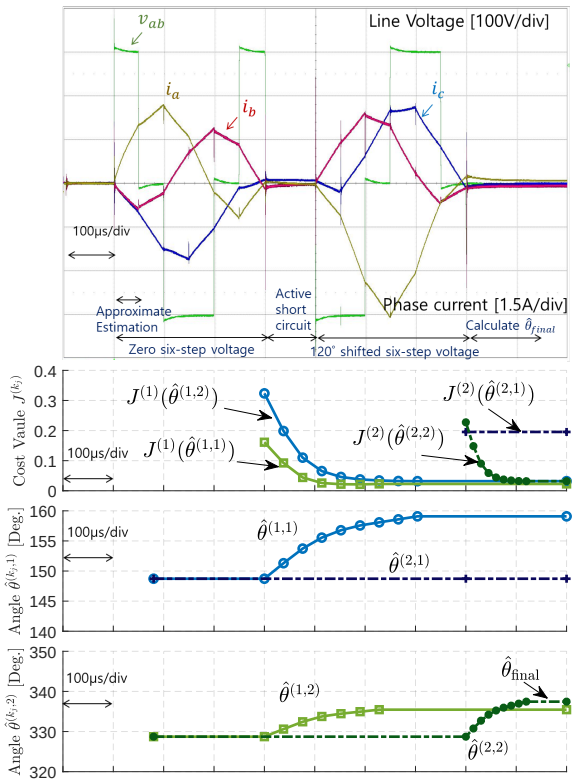


Fig. 16: The current response, six-step voltage injections, and cost value, and estimated angle when $\theta = 334.9°$.

lookup table (LUT), which is constructed offline as a function of the rotor angle and the time step k. Note that these two parts are not required to be executed simultaneously. The voltage injection/current acquisition part is synchronized with the sampling period, whereas the estimated-angle adaptation part can be executed independently as a background task. Therefore, the cost-function update does not need to be synchronized with every sampling time. Fig. 14 shows the flow chart of the proposed algorithm. The two parts execute without synchronization and are executed in parallel. The upper part represents the voltage injection process, where two predefined six-step voltage injections are applied independently of the estimated angle. An active-short state is inserted between the two six-step injections to stabilize the current waveform. The estimated-angle adaptation algorithm is executed based on the measured current data. The approximate estimation algorithm first determines an initial rotor angle, and Algorithm 2 is then performed after the six-step voltage injection is completed. Finally, the estimated rotor position is determined from the cost-function evaluation.

The estimated current was stored offline as LUT, a function of the rotor angle and discrete time step k. The data was sampled at $10°$ intervals. Taking into account the six-step voltage phases and $\alpha\beta$-axis current, four LUTs were constructed. The LUTs were constructed from experimental measurements to reduce the influence of unmodeled characteristics, such as multiple saliency and cross coupling effect. The construction process is summarized as follows.

- The $\alpha\beta$-axis current responses are measured at various

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rotor positions under the same voltage injection condition.

- The measured current-response data are first arranged according to the voltage-vector index and rotor position, and then smoothed using curve fitting.
- The fitted current response characteristics are stored in a LUT. It takes the discrete time step and rotor angle as inputs and outputs the corresponding $\alpha\beta$-axis current responses.

Fig. 15 - Fig. 18 show the experimental results obtained under different initial rotor positions. Each figure presents the two six-step voltage injections and the corresponding evolution of the cost function according to the gradient-based estimation algorithm. The voltage injection and current measurement are synchronized with the sampling period, whereas the gradient-based cost-function update can be executed separately as a background task.

Furthermore, the current waveforms show short-term oscillations immediately after the inverter switching state changes. These oscillations are mainly caused by common-mode noise generated during the switching transition and coupled into the current measurement signals.

Fig. 15 shows the phase current response under two six-step voltage injection and the corresponding cost value variation with the estimated rotor angle when $\theta = 155.4°$. First, a six-step voltage without phase shift was injected for $300\mu s$, followed by $100\mu s$ active short circuit to stabilize the current. Subsequently, a six-step voltage with a $120°$ phase shift was injected for an additional $300\,\mu$s. After the first voltage vector of the the six-step injection without phase shift was applied, the approximate initial angle estimation algorithm in Algorithm 1 was executed. Then, the resulting approximate estimate was used to initialize four angle estimates. After completion of the first six-step injection, Algorithm 2 was executed to compute two angle estimates, $\hat{\theta}^{(1,1)}$ and $\hat{\theta}^{(1,2)}$, by minimizing the corresponding cost function $J^{(1)}$. Afterward, the second six-step voltage injection with a $120°$ phase shift was performed, and two additional angle estimates were obtained by minimizing the cost function $J^{(2)}$. In this case, the difference between the two minimizers of $J^{(2)}$, $|J^{(2)}(\hat{\theta}^{(2,1)}) - J^{(2)}(\hat{\theta}^{(2,2)})|$ was 0.39, whereas the corresponding difference for $J^{(1)}$, $|J^{(1)}(\hat{\theta}^{(1,1)}) - J^{(1)}(\hat{\theta}^{(1,2)})|$ was 0.005. Therefore, according to the selection criteria defined in (27) and (28), the final estimated rotor angle was determined as $\hat{\theta}_{\text{final}} = \hat{\theta}^{(2,1)} = 158.1°$. Note that the number of iterations of the gradient-based optimization algorithm is not fixed. For example, in Fig. 15, $\hat{\theta}^{(2,1)}$ is updated only twice, whereas $\hat{\theta}^{(2,2)}$ is updated nine times. This difference is caused by the termination condition of the gradient-based optimization algorithm. The number of data points in each estimated position trajectory is not fixed because the gradient-based optimization is terminated when $\left|\nabla J^{(k_j)}\left(\hat{\theta}^{(k_j,i)}\right)\right| \leq \epsilon$ in Algorithm 2. Therefore, the number of plotted data points depends on how quickly each initial candidate satisfies the convergence condition.

Fig. 16 shows the current responses under the two six-step voltage injections when $\theta = 334.9°$. The rotor angle differs by approximately $180°$ from that in Fig. 15. Similar to the previous experiment, Algorithm 1 was first executed. After fully injecting the six-step voltage injection, Algorithm 2 was performed to compute four candidate angles by minimizing the cost functions $J^{(1)}$ and $J^{(2)}$. The differences between the two minimizers of each cost function were then evaluated. In this case, $|J^{(1)}(\hat{\theta}^{(1,1)}) - J^{(1)}(\hat{\theta}^{(1,2)})| = 0.009$ whereas $|J^{(2)}(\hat{\theta}^{(2,1)}) - J^{(2)}(\hat{\theta}^{(2,2)})| = 0.16$. Therefore, the final estimated rotor angle was determined as $\hat{\theta}_{\text{final}} = \hat{\theta}^{(2,2)} = 337.45°$.

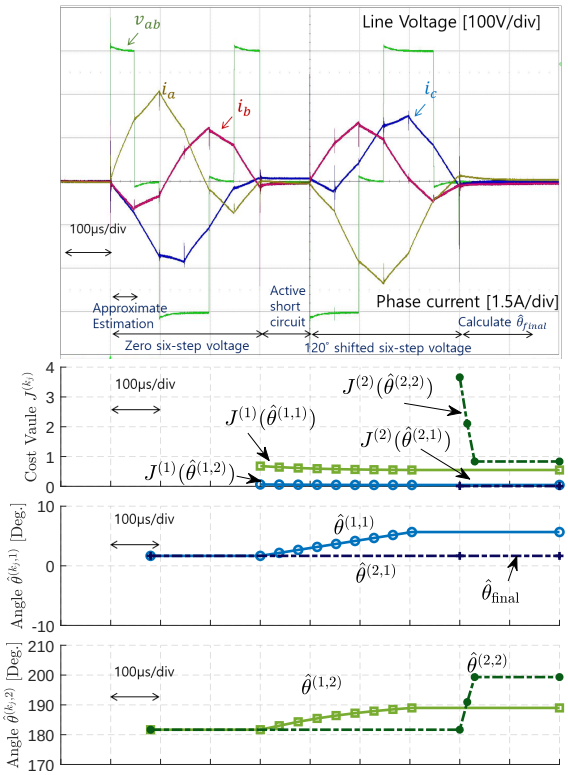


Fig. 17: The current response, six-step voltage injections, and cost value, and estimated position when $\theta = 5.1°$.

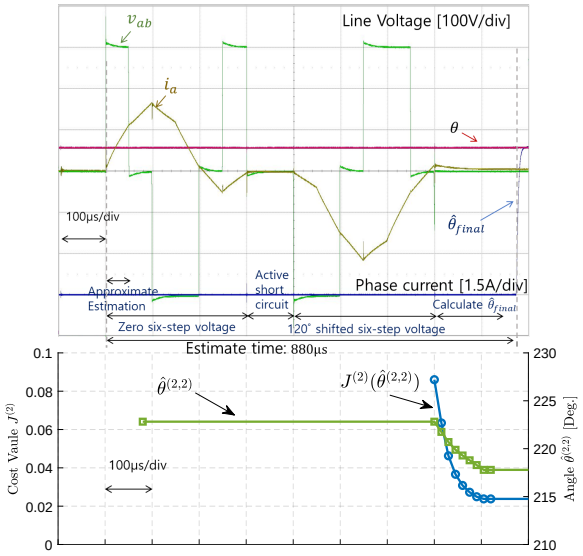


Fig. 18: The current response, six-step voltage injections, and cost value, and estimated position when $\theta = 215.9°$.

Similarly, Fig. 17 shows the current responses, the corre-

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sponding cost function values, and the four candidate angle estimates when the rotor position was $5.14°$. In this case, the difference between the global and local minima of $J^{(2)}$ was larger than that of $J^{(1)}$. Therefore, the final estimated rotor position was determined as $\hat{\theta}_{\text{final}} = \hat{\theta}^{(2,2)} = 1.68°$.

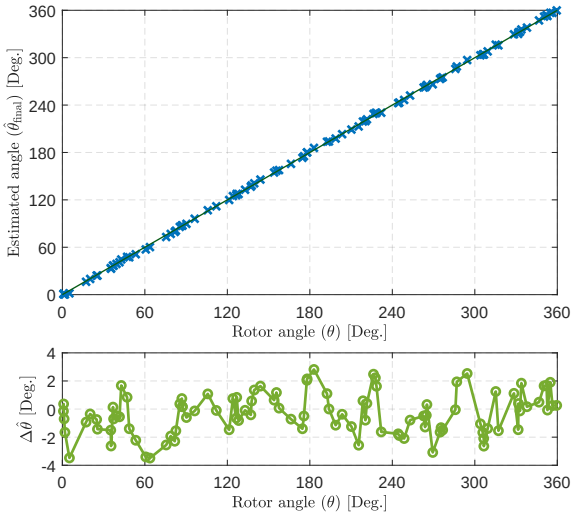


Fig. 19: Initial rotor position estimation under 100 random full-range tests.

Fig. 18 shows the line voltage, a phase current, the rotor position, and the final estimated position when $\theta = 215.9°$. It also shows the evolution of $\hat{\theta}^{(2,2)}$ under the gradient-based optimization and the corresponding variation of the cost function $J^{(2)}$. The cost function $J^{(2)}$ was minimized when $\hat{\theta}^{(2,2)}$ converged to approximately $217.79°$. Using (27) and (28), the final estimated rotor position was determined as $\hat{\theta}_{\text{final}} = \hat{\theta}^{(2,2)} = 217.79°$. Fig. 19 shows the experimental results obtained from 100 repeated initial position estimations over the full rotor position range from $0°$ to $360°$. The maximum estimation error remained within $\pm 4°$, demonstrating accurate and robust initial angle estimation across the entire operating range. While the six-step voltage injection generates transient d- and q-axis currents, the short injection duration prevents noticeable rotor acceleration, and the rotor shaft remains practically stationary. In the repeated experiments, no noticeable rotor vibration, speed buildup, or rotor position change was observed.

Table II compares the proposed method with conventional initial rotor position estimation methods. Compared with the conventional methods, the proposed method achieves a short convergence time of 950 μs while maintaining a maximum estimation error of $3.9°$. In addition, the proposed method requires only two voltage pulses and does not require an additional pulse for polarity checking. Although offline data are used, the proposed method enables parallel processing between the voltage injection and the estimation algorithm, which contributes to fast initial position estimation.

TABLE II: Comparison With Conventional Methods

Criteria	[19]	[27]	[30]	[26]	[23]	Proposed Method
Max. estimation time	30ms	10ms	200ms	1.5ms	12ms	**950μs**
Max. estimation error ($\Delta\theta$)	15°	3.9°	7.5°	3.9°	1°	**3.9°**
Voltage pulse number	6	16	3	2	3	**2**
Adaption algorithm	No	Yes	No	Yes	Yes	**Yes**
Additional Pulse for polarity check	No	Yes	No	No	No	**No**
Offline data needed	No	No	No	Yes	Yes	**Yes**
Parallel processing	No	No	No	No	No	**Yes**
Motor type	PM	PM	PM	SRM	SRM	**PM**
Calculation burden	low	low	low	medium	medium	**high**

VI. Conclusion

Using the two six-step voltage injection, a fast initial rotor position estimation method was developed. To obtain the initial rotor position from the current response, a cost function–based optimization framework was used and analytically investigated under both linear and nonlinear motor models. To improve estimation accuracy, six-step voltage injections without phase shift and with a 120° phase shift are applied, yielding two cost functions. To identify the unique global minimum, an approximate initial position and a gradient-based method are employed. The approximate rotor position was obtained using only the current response from the first switching state of the six-step voltage injection and used to initialize the gradient-based optimization. Due to the presence of local minima, the optimization is independently performed from two initial points separated by 180°. The final rotor position is determined by selecting the solution with the lower cost. Experimental results verify that the proposed method reliably estimates the rotor position over a wide range of initial conditions, achieving an accuracy within 4° in less than 1 ms.

Appendix

From (17), the expression can be simplified using trigonometric identities assuming that the rotor is at standstill, as follows:

$$K_t = \sum_{k=1}^{N_s} \left\| \begin{bmatrix} -\sin(2\theta + \Delta\theta)a_{ks} + \cos(2\theta + \Delta\theta)b_{ks} \\ \cos(2\theta + \Delta\theta)a_{ks} + \sin(2\theta + \Delta\theta)b_{ks} \end{bmatrix} \right\|^2$$

Thus, K_t is derived as $K_t = \sum_{k=1}^{N_s} a_{ks}^2 + b_{ks}^2$.

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